Product Introduction

The ZDH1112 is a power amplifier based on gallium arsenide (GaAs) HBT technology, operating from DC to 1000MHz. The ZDH1112 is designed for single-voltage supply from 3V to 9V. At a 5V supply and a 433MHz operating frequency, it can achieve an output power of 32dBm and an efficiency of 62%.

The ZDH1112 is packaged in a DFN2x2-8 package, offering excellent reliability and cost-effectiveness.

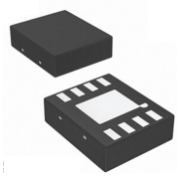
Typical application scenarios

- 315MHz, 433MHz ISM data transmission
- Walkie-talkies
- Model aircraft remote control
- Meter reading systems

Absolute Maximum Ratings

Parameter	Value
Storage Temperature	-65°C~+150°C
Operating Temperature	-55°C~+105°C
Supply Voltage (Vcc)	+13V
Input Power (RFIN)	+20dBm

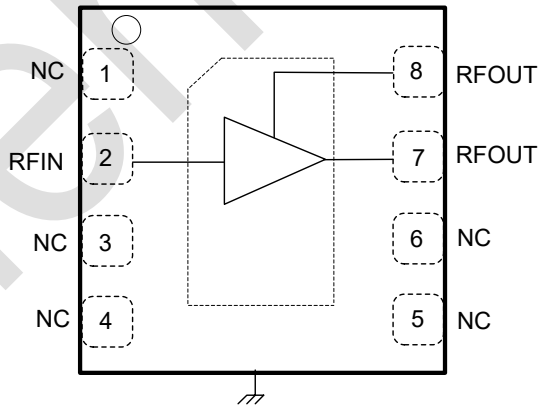
Features



- 3V-9V single voltage supply, typical current 535mA @ 5V
- Small signal gain: 21dB @ 433MHz, VCC=5V
- Typical OIP3: 43dBm @ 433MHz, VCC=5V
- Efficiency: >62% @ 433MHz, VCC=5V
- Input/output 50Ω impedance matching
- Green lead-free 8-pin DFN2x2 package

This product complies with all relevant regulations and is halogen-free.

Pin diagram (Top View)



PIN No.	Pin Name	Description
1,3,4,5,6	NC	Floating or grounded
2	RFIN	RF Input
7,8	RFOUT	RF Output
9	EPAD	GND

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Electrical parameters

1. Test conditions: VCC=5V, Temp= +25°C, 160MHz~180MHz application circuit, 50Ω test system.

Parameter	Value			Unit
Frequency	160	169	180	MHz
Small-signal gain	25	25	25	dB
Input return loss (S11)	-26	-29	-25	dB
OIP3	43	43	43	dBm
Output power (Pout)	32	32	32	dBm
Input power (Pin)	15	14	15	dBm
Quiescent current (Icq)	430	430	430	mA
Operating current (Icc)	500	504	517	mA
Efficiency	63	63	61	%

2. Test conditions: VCC=5V, Temp= +25°C, 400MHz~470MHz application circuit, 50Ω test system.

Parameter	Value			Unit
Frequency	400	433	470	MHz
Small-signal gain	22	21	20	dB
Input return loss (S11)	-15	-17	-14	dB
OIP3	44	43	42	dBm
Output power (Pout)	32	32	32	dBm
Input power (Pin)	15	15	15	dBm
Quiescent current (Icq)	430	430	430	mA
Operating current (Icc)	513	535	558	mA
Efficiency	68	62	63	%

3. Test conditions: VCC=5V, Temp= +25°C, 868MHz~915MHz application circuit, 50Ω test system.

Parameter	Value			Unit
Frequency	868	891	915	MHz
Small-signal gain	15	15	15	dB
Input return loss (S11)	-13	-13	-14	dB
OIP3	40	40	40	dBm
Output power (Pout)	31	31	31	dBm
Input power (Pin)	19	19	19	dBm
Quiescent current (Icq)	432	430	430	mA
Operating current (Icc)	528	526	525	mA
Efficiency	47	49	49	%

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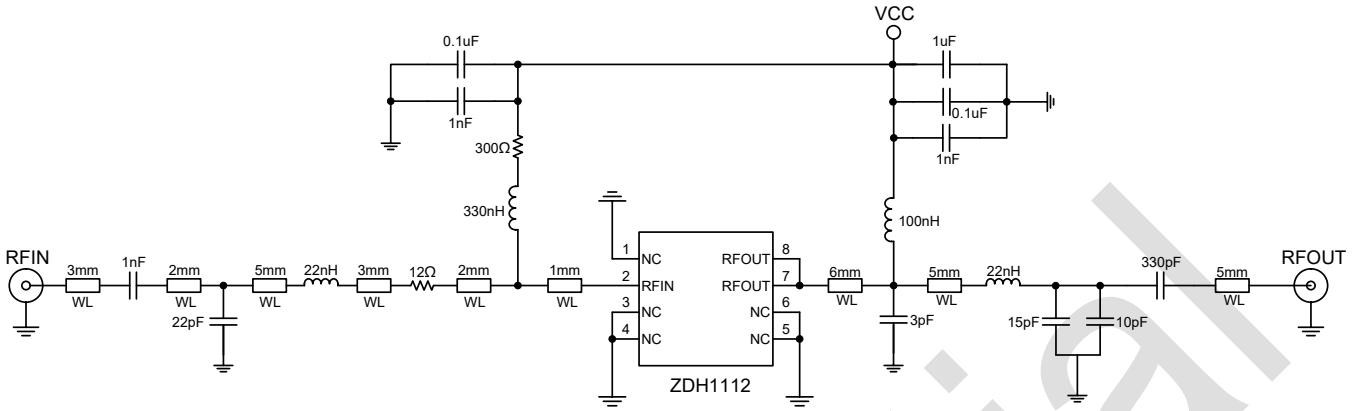
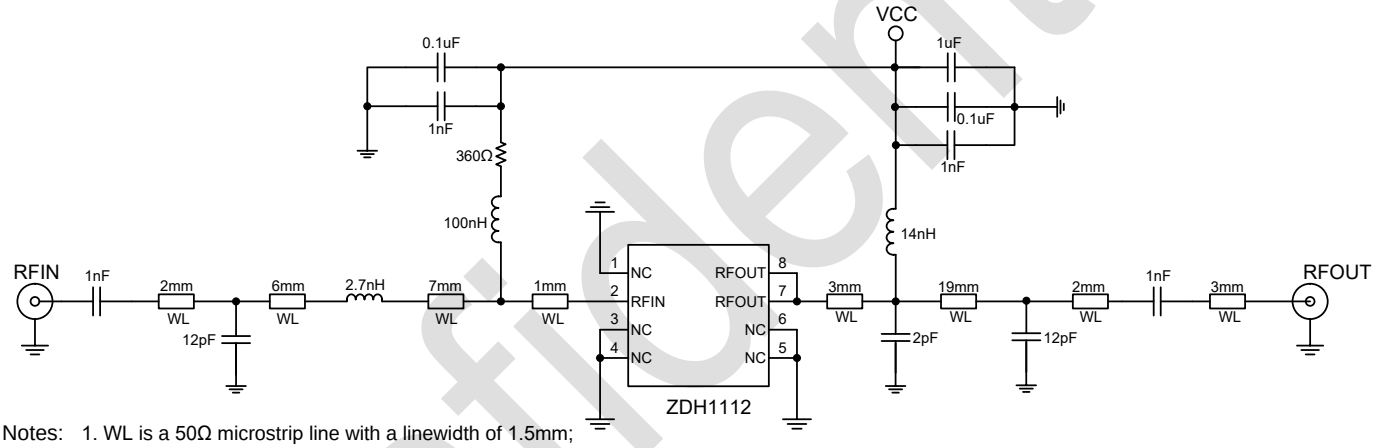
Application circuit (160MHz~180MHz, VCC=5V)



Notes: 1. WL is a 50Ω microstrip line with a linewidth of 1.5mm;
2. The PCB material has a dielectric constant of 4.4 and a thickness of 0.8mm.

Application circuit (400MHz~470MHz, VCC=5V)



Notes: 1. WL is a 50Ω microstrip line with a linewidth of 1.5mm;
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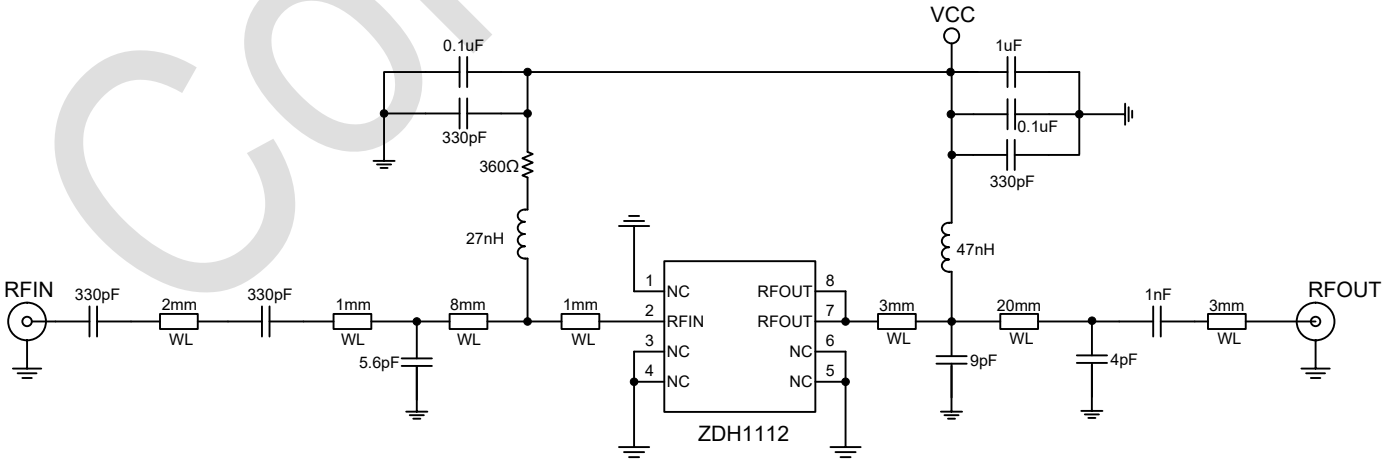
Application circuit (868MHz~915MHz, VCC=5V)



Notes: 1. WL is a 50Ω microstrip line with a line width of 1.5mm;
2. The PCB board has a dielectric constant of 4.4 and a board thickness of 0.8mm.

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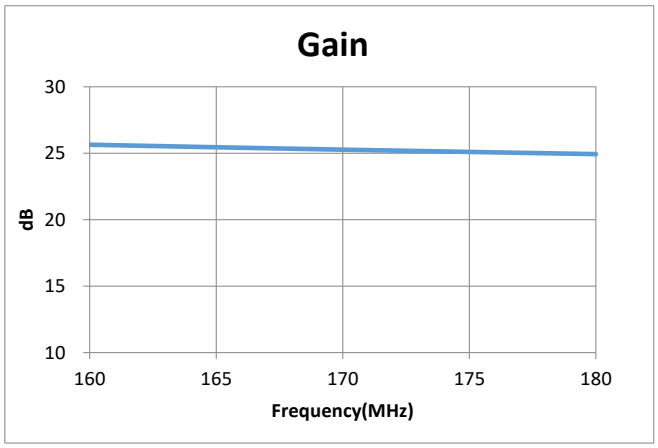
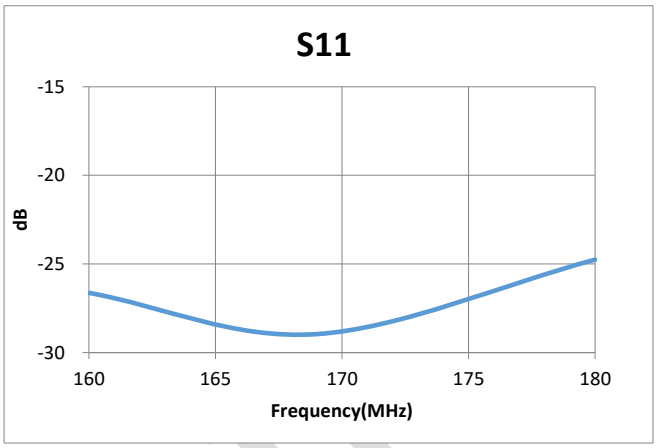
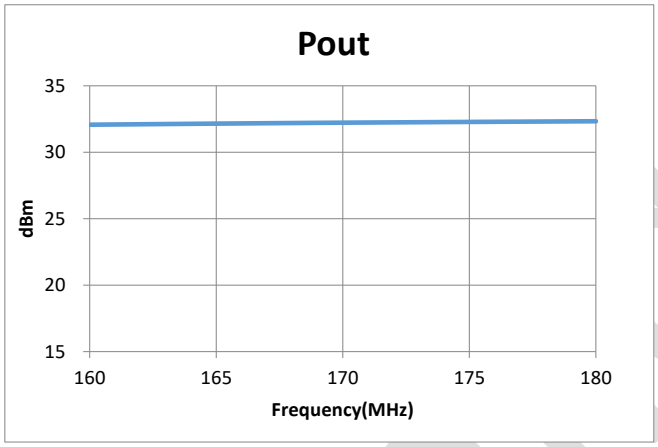
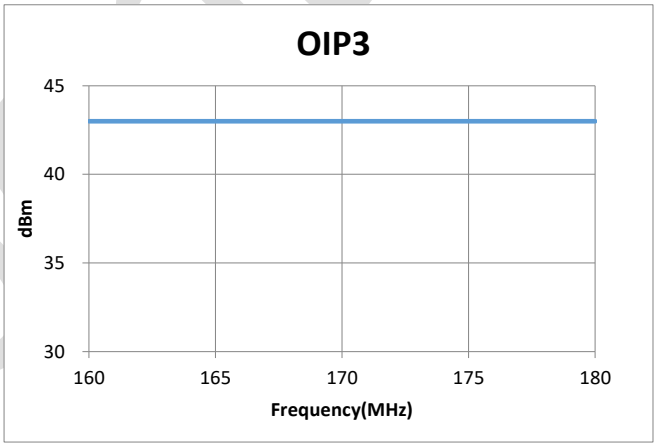
Typical performance curves

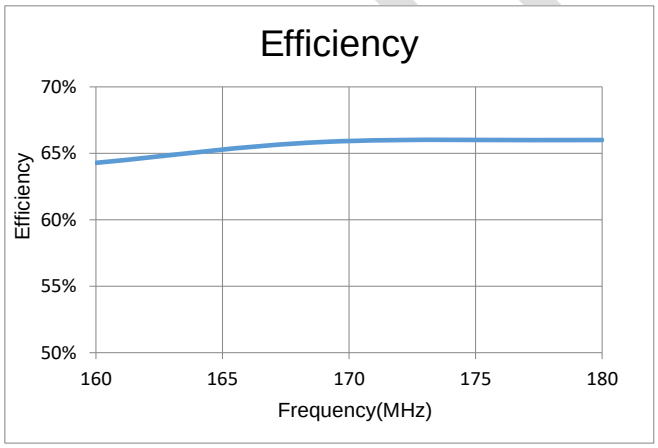
1. Test conditions: VCC=5V, Temp= +25°C, 160MHz~180MHz application circuit, 50Ω test system.





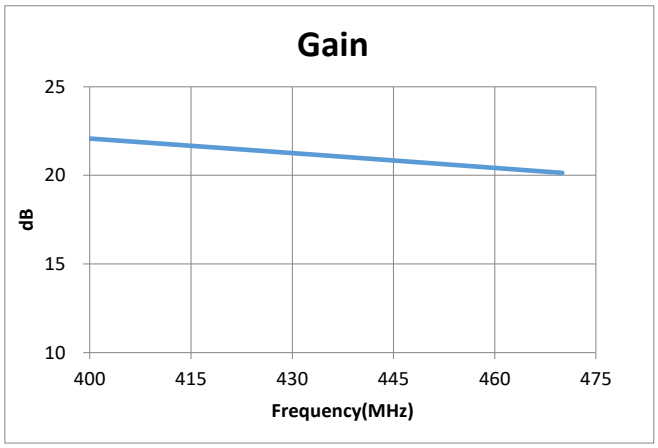


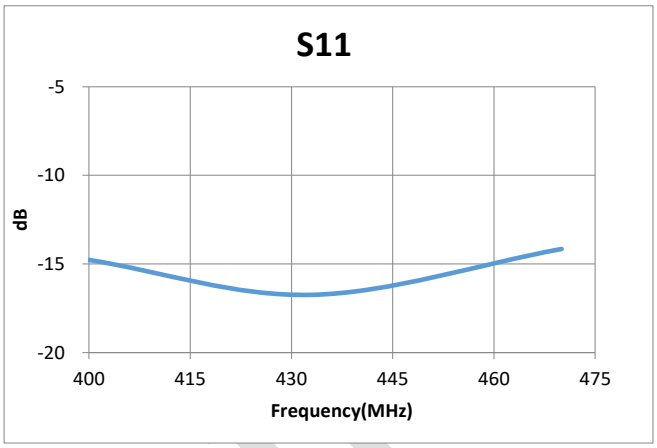


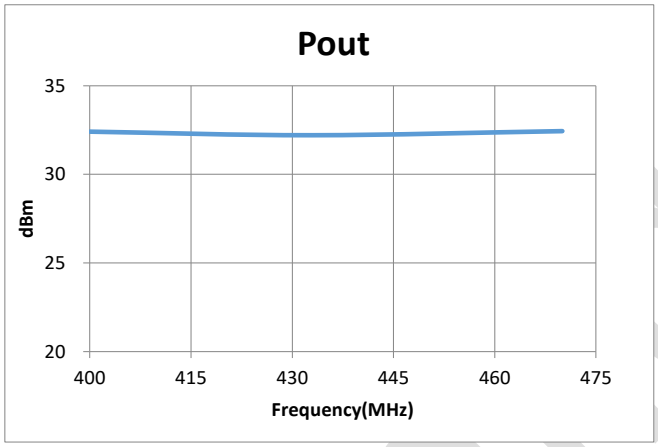


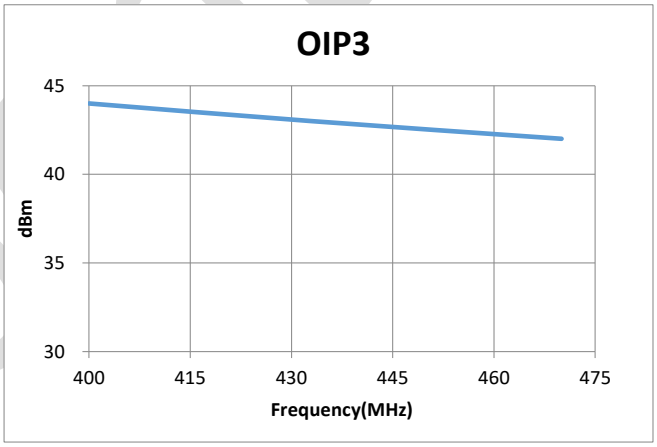
Typical performance curves

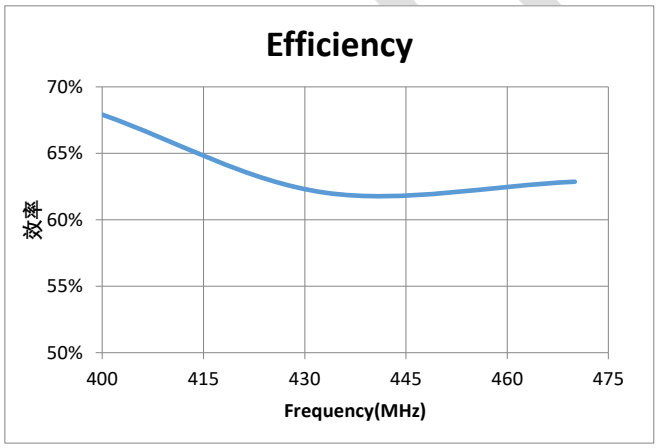
2. Test conditions: VCC=5V, Temp= +25°C, 400MHz~470MHz application circuit, 50Ω test system.





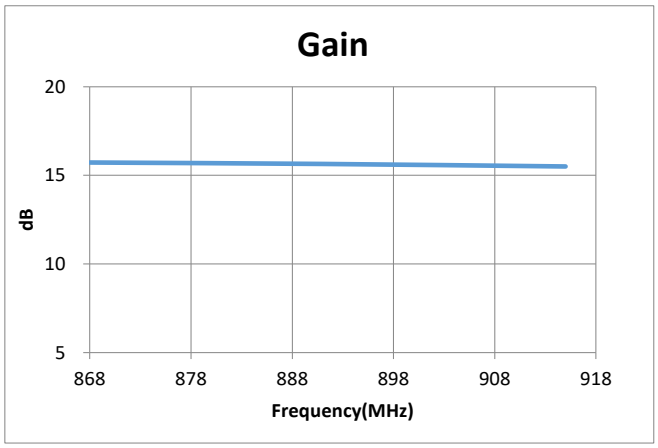


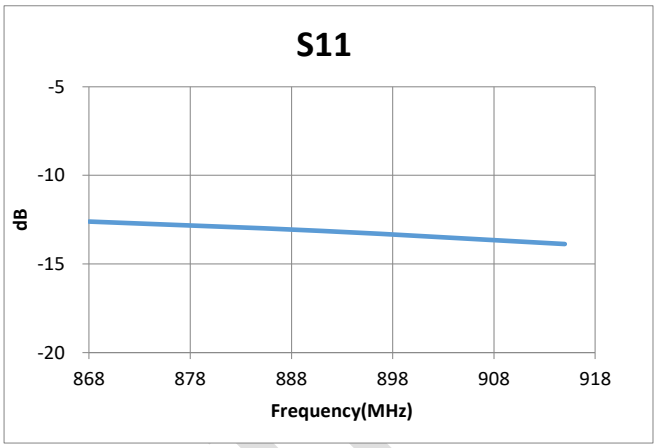


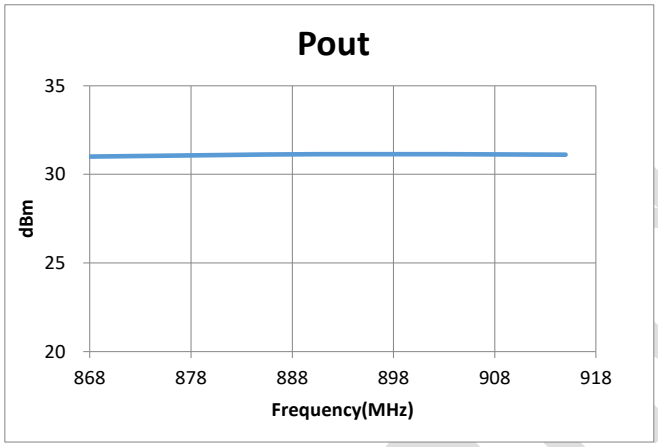


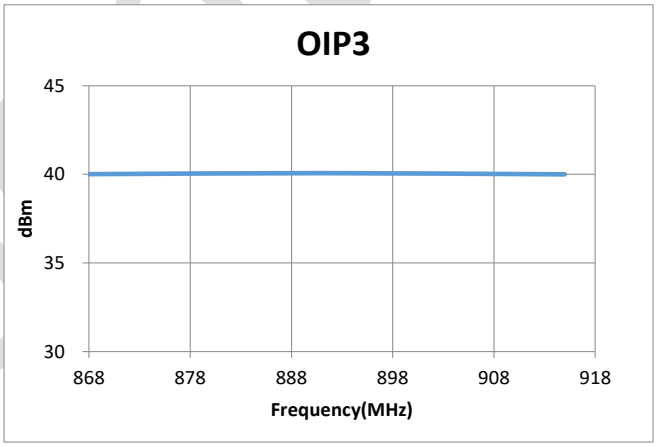
Typical performance curves

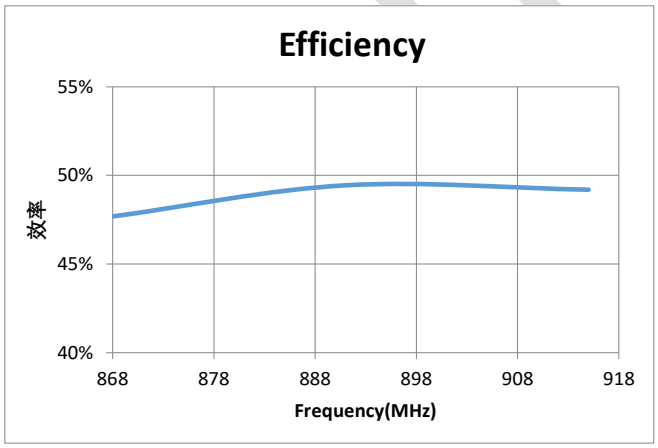
3. Test conditions: VCC=5V, Temp= +25°C, 868MHz~915MHz application circuit, 50Ω test system.



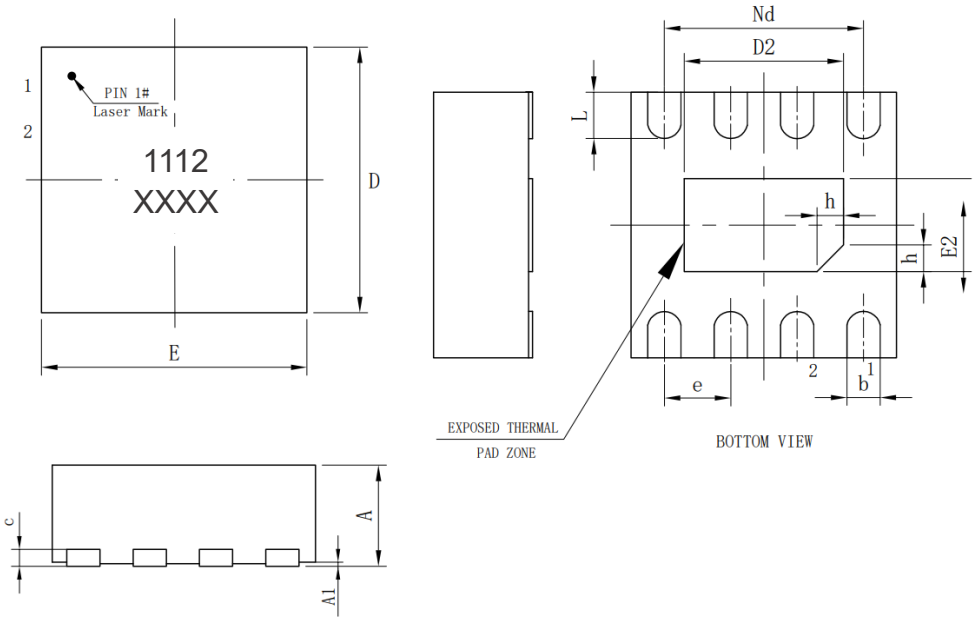








Package size diagram



SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	0. 70	0.75	0. 80
A1	—	0.02	0. 05
b	0. 18	0.25	0. 30
c	0. 18	0.20	0. 25
D	1. 90	2. 00	2. 10
D2	1. 10	1. 20	1. 30
e	0. 50BSC		
Nd	1. 50BSC		
E	1. 90	2. 00	2. 10
E2	0. 60	0. 70	0. 80
L	0. 30	0.35	0. 40
h	0. 15	0. 20	0. 25
载体尺寸（mil）	63X39		

Ordering Information

Model	Marking	Packaging
ZDH1112	1112	DFN2x2-8

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